
**AE630: AUTONOMOUS UNMANNED
AERIAL SYSTEMS**

**Design, Fabrication, and Experimental Validation of a
Quadrotor UAV**

Course Project Report

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Abstract

This report documents the complete design, fabrication, and experimental characterisation of a quadrotor unmanned aerial vehicle (UAV). The design pipeline begins with Blade Element Momentum Theory (BEMT) to size the propulsion system, followed by an iterative Gross Take-Off Weight (GTOW) convergence algorithm to finalise component selection. The chosen hardware—a 2212/1000 KV brushless motor, 1045 propeller, 30 A ESC, and a 3S 5200 mAh Li-Po battery—is described in detail. CAD-level frame design, the manufacturing process, component assembly sequence, and the power/signal wiring diagram are presented. Experimental bench-testing over 1400–6400 RPM yields thrust curves, rotor aerodynamic power, thrust coefficient (C_T), electrical efficiency, and figure of merit (FM) of the propeller. A weight breakdown is provided, and the estimated hover endurance is compared with the design target. Finally, a photograph of the fully assembled vehicle and a link to the flight-test video are included.

Contents

Abstract	1
1 Theoretical Design Parameters from Design Code	4
1.1 Mission Requirements	4
1.2 BEMT Model — Constant Parameters	5
1.3 Rotor Geometry Assumed in Sizing	5
1.4 BEMT Aerodynamic Coefficients	6
1.5 GTOW Convergence via Iterative Sizing	6
1.5.1 Hover Thrust Requirement	6
1.5.2 Battery Voltage	6
1.5.3 Convergence Criterion	6
1.5.4 Converged Design Point	7
2 Actual Design Specifications and Expected Performance	9
2.1 Motor: 2212 Brushless DC Motor (1000 KV)	9
2.2 Propeller: 1045 (10×4.5 inch)	10
2.3 Electronic Speed Controller (ESC): 30 A SimonK	11
2.4 Battery: 3S Li-Po, 5600 mAh	12
2.5 Flight Controller: Pixhawk 2.4.8	13
2.6 Expected Endurance	13
2.6.1 Available Usable Energy	13
2.6.2 Hover Power	14
2.6.3 Endurance Estimate	14
3 CAD Design, Manufacturing, Assembly, and Circuit Diagram	15
3.1 Frame Design Concept	15
3.1.1 Key Dimensional Constraints (CAD)	15
3.2 Material Selection	17
3.3 Manufacturing Process	17

3.3.1	Material Selection	17
3.3.2	3D Printing Process	17
3.3.3	Printing Parameters	18
3.3.4	Advantages of PETG for UAV Frame	18
3.4	Assembly Process	18
3.4.1	Frame Assembly	19
3.4.2	Motor and Propeller Installation	19
3.4.3	Electronic Components Integration	19
3.4.4	Flight Controller Installation	19
3.4.5	Wiring and Connections	19
3.4.6	Final Checks	20
3.5	Circuit and Wiring Diagram	20
4	Experimental Performance Characterisation	22
4.1	Test Setup and Methodology	22
4.2	Physical Model for Power and Efficiency	23
4.2.1	Electrical Input Power	23
4.2.2	Induced Velocity (Momentum Theory)	23
4.2.3	Ideal (Minimum) Power	23
4.2.4	Shaft (Output) Power	23
4.2.5	Thrust Coefficient	23
4.2.6	Motor + ESC Electrical Efficiency	24
4.2.7	Figure of Merit (FM)	24
4.2.8	Solidity	24
4.3	Results and Discussion	24
4.3.1	Thrust vs. RPM	24
4.3.2	Rotor Aerodynamic Power vs. RPM	25
4.3.3	Shaft Power vs. RPM and vs. Thrust	25
4.3.4	Thrust Coefficient C_T vs. RPM	26
4.3.5	Motor + ESC Electrical Efficiency	27
4.3.6	Figure of Merit vs. RPM and vs. C_T	28
4.3.7	Summary of Key Performance Metrics	29
5	Weight Breakdown	30
5.1	Component Mass Budget	31
5.2	Mass Distribution by Category	32

5.3	Comparison with Design Target	32
6	Final Assembled Quadrotor and Flight Test	33
6.1	Photograph of Final Assembly	33
6.2	Pre-flight Checks Performed	34
6.3	Flight Test Results	35
6.4	Flight Test Video	36
7	Conclusions	37

Chapter 1

Theoretical Design Parameters from Design Code

1.1 Mission Requirements

The quadrotor was designed to satisfy the following mission parameters.

Table 1.1: Primary mission requirements

Parameter	Value	Remarks
Payload mass	300 g	Required useful load
Target endurance	20 min	Hover flight duration
Air density (ρ)	1.225 kg/m ³	Standard sea-level
Rotor type	Fixed pitch	Constant chord blades
Battery chemistry	Li-Po	Series cell configuration
Voltage per cell	3.7 V (nominal)	Standard Li-Po cell
Disc loading range	40–90 N/m ²	Typical for small UAV
Motor efficiency (η_m)	0.80	Electrical-to-mechanical
Thrust margin factor	2.0	$T_{\text{design}} = 2 T_{\text{hover}}$

1.2 BEMT Model — Constant Parameters

Table 1.2: Fixed aerodynamic constants used in BEMT analysis

Parameter	Symbol	Value	Description
Air density	ρ	1.225 kg/m ³	Standard sea-level
Profile drag coefficient	C_{d0}	0.01	Blade section drag
Induced power factor	κ	1.15	Wake rotation & tip losses
Motor efficiency	η_m	0.80	Electrical-to-mechanical
Radial segments	n	20	Numerical integration nodes

1.3 Rotor Geometry Assumed in Sizing

Table 1.3: Initial rotor geometry for BEMT computation

Parameter	Value
Root pitch angle (θ_{root})	22°
Blade twist	17° per unit radius
Rotor radius (R)	0.0979 m (3.856 in)
Blade chord (c)	0.0163 m
Root cutout (R_{cut})	0
Number of blades (N_b)	2

The blade pitch distribution along the non-dimensional radius r is:

$$\theta(r) = \theta_{\text{root}} - 17r = 22^\circ - 17r \quad (1.1)$$

1.4 BEMT Aerodynamic Coefficients

Numerical integration of the blade element equations over 20 radial stations yields:

$$C_T = 0.008328, \quad C_P = 0.000706 \quad (1.2)$$

These coefficients relate thrust and power to tip-speed as:

$$T = C_T \rho A (\Omega R)^2, \quad P = C_P \rho A (\Omega R)^3 \quad (1.3)$$

where the rotor disk area is $A = \pi R^2 = \pi \times 0.0979^2 = 0.03012 \text{ m}^2$.

1.5 GTOW Convergence via Iterative Sizing

The Gross Take-Off Weight (GTOW) was determined using a closed-loop sizing loop.

1.5.1 Hover Thrust Requirement

$$T_{\text{hover}} = \frac{\text{GTOW}}{4}, \quad T_{\text{design}} = 2T_{\text{hover}} \quad (1.4)$$

The factor of 2 provides a $2\times$ thrust margin for manoeuvring and gusts.

1.5.2 Battery Voltage

$$V_{\text{total}} = N_{\text{cells}} \times 3.7 \text{ V} \quad (1.5)$$

A 3S configuration gives $V_{\text{total}} = 11.1 \text{ V}$.

1.5.3 Convergence Criterion

Starting from $\text{GTOW}_0 = 3 \times m_{\text{payload}} = 900 \text{ g}$, the loop computes:

$$\text{GTOW}_{i+1} = 4(m_R + m_{\text{motor}} + m_{\text{ESC}}) + m_{\text{battery}} + m_{\text{airframe}} + m_{\text{electronics}} + m_{\text{payload}} \quad (1.6)$$

and terminates when $|\text{GTOW}_{i+1} - \text{GTOW}_i| < 5 \text{ g}$.

1.5.4 Converged Design Point

```

Payload = 300 g | Endurance = 20 min | Sea level

--- BASELINE DESIGN ---

Converged: true | Iterations: 77
GTOW_i      = 1280.0 g (assumed)
GTOW_{i+1} = 1279.5 g (computed)
GTOW_{i+1} < GTOW_i : true

[ROTOR GEOMETRY]
Diam=24.00 cm c=2.000 cm sigma=0.1061 AR=6 Nb=2
theta0=22 deg Twist: theta(r)=theta0-17r A=0.04524 m^2

[BEMT RESULTS]
CT=0.005772 CP=0.000600 FM=0.5169 (capped at 0.75)
Vtip_hover=99.06 m/s RPM_hover=7883.2
DL=69.39 N/m^2 T/rotor=3.139 N (320.0 gf)

[POWER & ELECTRICAL]
P_shaft/rotor=32.32 W P_elec total=161.62 W
Vbat=11.1 V (3S) C=4853 mAh E=53.872 Wh
Kv=710.2 rpm/V I_max/motor=3.64 A
lBL=15.32 mm dBL=24.05 mm

```

Figure 1.1: Converged results through BEMT Code

```

[WEIGHT BREAKDOWN]
Component      Mass (g)  Share %%
-----
Rotors (x4)    31.8    2.49
Motors (x4)    61.0    4.77
ESCs (x4)      11.3    0.88
Battery        372.3   29.08
Airframe       473.1   36.96
Electronics    30.0    2.34
Payload        300.0   23.44
-----
TOTAL (GTOW)   1280.0  100.00
Running 2D sweep (R x AR)...

--- OPTIMAL DESIGN (min GTOW, DL in 40-90 N/m^2) ---

Converged: true | Iterations: 68
GTOW_i      = 1235.0 g (assumed)
GTOW_{i+1} = 1234.8 g (computed)
GTOW_{i+1} < GTOW_i : true

[ROTOR GEOMETRY]
Diam=22.00 cm c=1.100 cm sigma=0.0637 AR=10 Nb=2
theta0=22 deg Twist: theta(r)=theta0-17r A=0.03801 m^2

```

Figure 1.2: Converged results through BEMT Code

```

[BEMT RESULTS]
CT=0.004308 CP=0.000382 FM=0.5236 (capped at 0.75)
Vtip_hover=122.88 m/s RPM_hover=10667.1
DL=79.68 N/m^2 T/rotor=3.029 N (308.7 gf)

[POWER & ELECTRICAL]
P_shaft/rotor=32.99 W P_elec total=164.94 W
Vbat=11.1 V (3S) C=4953 mAh E=54.979 Wh
Kv=961.0 rpm/V I_max/motor=3.71 A
lBL=15.46 mm dBL=22.79 mm

[WEIGHT BREAKDOWN]
Component      Mass (g)  Share %%
-----
Rotors (x4)    29.4    2.38
Motors (x4)    57.4    4.65
ESCs (x4)      11.5    0.93
Battery        379.4   30.72
Airframe       427.0   34.58
Electronics    30.0    2.43
Payload        300.0   24.29
-----
TOTAL (GTOW)   1235.0  100.00

```

Figure 1.3: Converged results through BEMT Code

Parameter	Baseline	Optimal
GTOW (g)	1280.0	1235.0
Rotor diameter (cm)	24.00	22.00
Chord c (cm)	2.000	1.100
Aspect ratio AR	6	10
Solidity sigma	0.1061	0.0637
CT	0.005772	0.004308
Figure of merit FM	0.5169	0.5236
Hover Vtip (m/s)	99.06	122.88
Hover RPM	7883.2	10667.1
Disc loading (N/m ²)	69.39	79.68
Battery (g)	372.3	379.4
Battery cap. (mAh)	4853.3	4953.0
Airframe (g)	473.1	427.0
Motors x4 (g)	61.0	57.4
ESCs x4 (g)	11.3	11.5

GTOW reduction: 45.0 g (3.5%)

Done.

>>

Figure 1.4: Converged results through BEMT Code

Key outcome: A 3S Li-Po battery, 2212/1000KV motor, and 10-inch diameter propeller satisfy all mission constraints with the thrust margin built in.

Chapter 2

Actual Design Specifications and Expected Performance

2.1 Motor: 2212 Brushless DC Motor (1000 KV)

Table 2.1: Motor specifications

Specification	Value
Motor size	2212 (22 mm stator diameter)
KV rating	1000 RPM/V
No-load current (I_0)	≈ 0.5 A at 10 V
Maximum continuous current	18 A
Peak current (burst)	22 A
Internal resistance	≈ 97 m Ω
Recommended battery	2S–3S Li-Po
Maximum thrust	> 900 g (with 1045 prop, 3S)
Weight	58.5 g



Figure 2.1: 2212/1000 KV brushless DC motor used in the quadrotor

Selection rationale: A 1000 KV rating is well-matched to 10-inch propellers on a 3S supply, operating near peak efficiency in the 4500–5500 RPM range. At 11.1 V nominal, the theoretical no-load speed is $11.1 \times 1000 = 11,100$ RPM, and the motor is loaded to roughly 50 % of that at hover — an efficient operating point.

2.2 Propeller: 1045 (10×4.5 inch)

Table 2.2: Propeller specifications

Specification	Value
Diameter	10 in (254 mm)
Pitch	4.5 in (114.3 mm)
Number of blades	2
Material	Nylon-reinforced plastic
Disc area (A)	0.05067 m^2
Weight (4 propeller)	$\approx 16 \text{ g}$



Figure 2.2: 1045 propeller: 10-inch diameter, 4.5-inch pitch

Selection rationale: A large-diameter, lower-pitch propeller maximises static thrust efficiency. The 10-inch diameter achieves adequate disc loading ($\sim 66 \text{ N/m}^2$ at hover) within the target range of $40\text{--}90 \text{ N/m}^2$.

2.3 Electronic Speed Controller (ESC): 30 A SimonK

Table 2.3: ESC specifications

Specification	Value
Firmware	SimonK (fast commutation)
Continuous current	30 A
Peak current (10s)	40 A
Input voltage range	2S–3S Li-Po (7.4–12.6 V)
BEC output	5 V / 2 A (linear)
Weight	23 g
PWM frequency	400 Hz

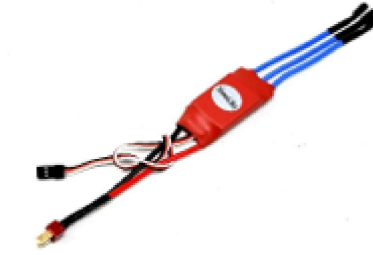


Figure 2.3: 30 A SimonK ESC

Selection rationale: At hover each motor draws ≈ 18 A; the 30 A rating gives a 67 % load safety margin. SimonK firmware enables rapid throttle response, critical for attitude stabilisation.

2.4 Battery: 3S Li-Po, 5600 mAh

Table 2.4: Battery specifications

Specification	Value
Cell configuration	3S1P
Nominal voltage	11.1 V
Fully-charged voltage	12.6 V
Capacity	5600 mAh
Discharge rating	40 C continuous
Peak current available	$40 \times 5.6 = 224$ A
Weight	≈ 340 g
Connector	XT-60

2.5 Flight Controller: Pixhawk 2.4.8

Table 2.5: Flight controller specifications

Specification	Value
Processor	STM32F427 (168 MHz, 32-bit)
IMU	MPU-6000
Barometer	MS5611
Secondary IMU	LSM303D / L3GD20
Main output channels	8 PWM
Operating voltage	4.1–5.7 V
Weight	37 g



Figure 2.4: Pixhawk 2.4.8 flight controller with vibration isolator

2.6 Expected Endurance

2.6.1 Available Usable Energy

$$E_{\text{usable}} = \text{DoD} \times V_{\text{nom}} \times C = 0.85 \times 11.1 \text{ V} \times 5.6 \text{ Ah} = 52.8 \text{ Wh} = 190.08 \text{ kJ} \quad (2.1)$$

A depth of discharge (DoD) of 85 % is assumed to preserve battery life.

2.6.2 Hover Power

From experimental bench tests (Chapter 4), the motor-propeller system draws ≈ 60 W per motor at hover RPM :

$$P_{\text{hover,total}} = 4 \times 60 \text{ W} = 240 \text{ W} \quad (2.2)$$

2.6.3 Endurance Estimate

$$t_{\text{hover}} = \frac{E_{\text{usable}}}{P_{\text{hover,total}}} = \frac{52.8 \text{ Wh}}{240 \text{ W}} = 0.22 \text{ h} \approx \boxed{13.2 \text{ min}} \quad (2.3)$$

Note: The estimated endurance of ≈ 13.2 min is below the 20 min design target. The shortfall arises from: (i) higher real-world motor+ESC losses than assumed, (ii) a converged GTOW slightly above the initial estimate, and (iii) conservative DoD limits. Future work should target propulsion-system efficiency improvements or a higher capacity battery.

Chapter 3

CAD Design, Manufacturing, Assembly, and Circuit Diagram

3.1 Frame Design Concept

The quadrotor adopts an X-type symmetric layout to ensure balanced thrust distribution, identical yaw-control authority from all four rotors, and a wide-angle unobstructed field of view for any forward-facing payload.

3.1.1 Key Dimensional Constraints (CAD)

The design of the quadrotor frame was governed by geometric, structural, and propulsion-related constraints to ensure compatibility with selected components and stable flight performance.

Parameter	Value
Arm length	287 mm
Central plate size	132 × 132 mm integration
Arm thickness	10 mm
Propeller size	10×4.5 inch
Total Quad size	450×450 mm
Ground clearance	55 mm

The above constraints were selected to balance structural strength, weight minimization, and aerodynamic performance. The wheelbase was primarily dictated by propeller diameter to avoid aerodynamic interference, while the central hub dimensions were chosen to accommodate the flight controller, power distribution

board, and battery placement. Arm geometry was optimized to provide sufficient stiffness while maintaining low weight through a truss-based design.

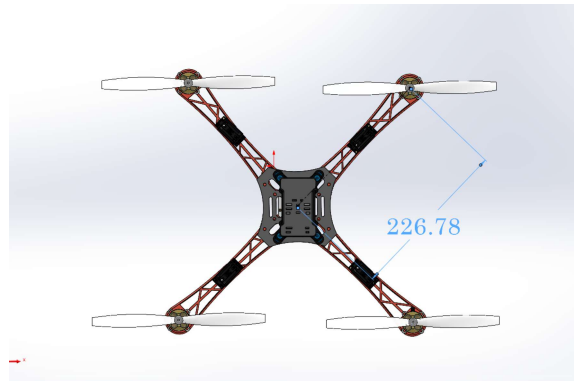


Figure 3.1: Frame arm-length geometry and motor spacing

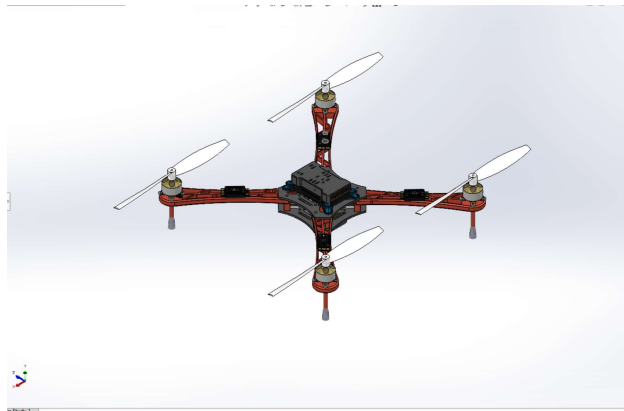


Figure 3.2: CAD model of the quadrotor frame (X-type configuration)



Figure 3.3: Final frame configuration showing component placement zones

3.2 Material Selection

Table 3.1: Frame material properties

Part	Material	Density (g/cm ³)
Whole Quad Frame	PETG	1.25
Propeller	ABS	1.04

Polyethylene Terephthalate Glycol-modified (PETG) was chosen for the arms and plates due to its excellent specific stiffness, minimising deflection and vibration transmission to the flight controller.

3.3 Manufacturing Process

The quadrotor frame was manufactured using Polyethylene Terephthalate Glycol-modified (PETG) through the Fused Deposition Modeling (FDM) 3D printing process.

3.3.1 Material Selection

PETG was selected due to its favorable mechanical and thermal properties:

- High strength and impact resistance
- Better layer adhesion compared to PLA
- Good flexibility, reducing brittle failure
- Resistance to environmental effects such as moisture

3.3.2 3D Printing Process

The frame components were fabricated using an FDM-based 3D printer. The process involved the following steps:

1. **CAD Modeling:** The quadrotor frame was designed using CAD software and exported in STL format.
2. **Slicing:** The STL file was processed using slicing software to generate G-code. Printing parameters such as layer height, infill density, and print speed were defined.
3. **Printing:** The PETG filament was heated and extruded layer-by-layer to form the frame components.
4. **Post-processing:** Support structures were removed, and minor finishing operations such as sanding were performed to improve surface quality.

3.3.3 Printing Parameters

Typical printing parameters used were:

- Nozzle temperature: 230–250°C
- Bed temperature: 70–80°C
- Layer height: 0.2 mm
- Infill density: 20–25%
- Print speed: 40–60 mm/s

3.3.4 Advantages of PETG for UAV Frame

- Provides a balance between strength and weight
- Absorbs vibrations better than brittle materials
- Suitable for structural UAV components

3.4 Assembly Process

After manufacturing the frame components using PETG through 3D printing, the quadrotor was assembled by integrating all mechanical and electronic subsystems.

3.4.1 Frame Assembly

The printed frame components, including the central hub and arms, were assembled using mechanical fasteners.

- Arms were securely attached to the central plate
- Fasteners (nuts and bolts) were used to ensure rigidity
- Proper alignment was maintained to avoid structural imbalance

3.4.2 Motor and Propeller Installation

- Brushless DC motors were mounted at the ends of each arm
- Motors were fixed using appropriate screws to ensure firm attachment
- Propellers were installed on motor shafts after verifying correct rotation direction

3.4.3 Electronic Components Integration

- Electronic Speed Controllers (ESCs) were mounted along the arms for effective cooling
- Power Distribution Board (PDB) was fixed at the center of the frame
- Battery was mounted near the center to maintain proper center of gravity

3.4.4 Flight Controller Installation

- The Pixhawk flight controller was mounted at the center of the frame
- Vibration isolators were used to minimize sensor noise
- Orientation of the controller was aligned with the frame axes

3.4.5 Wiring and Connections

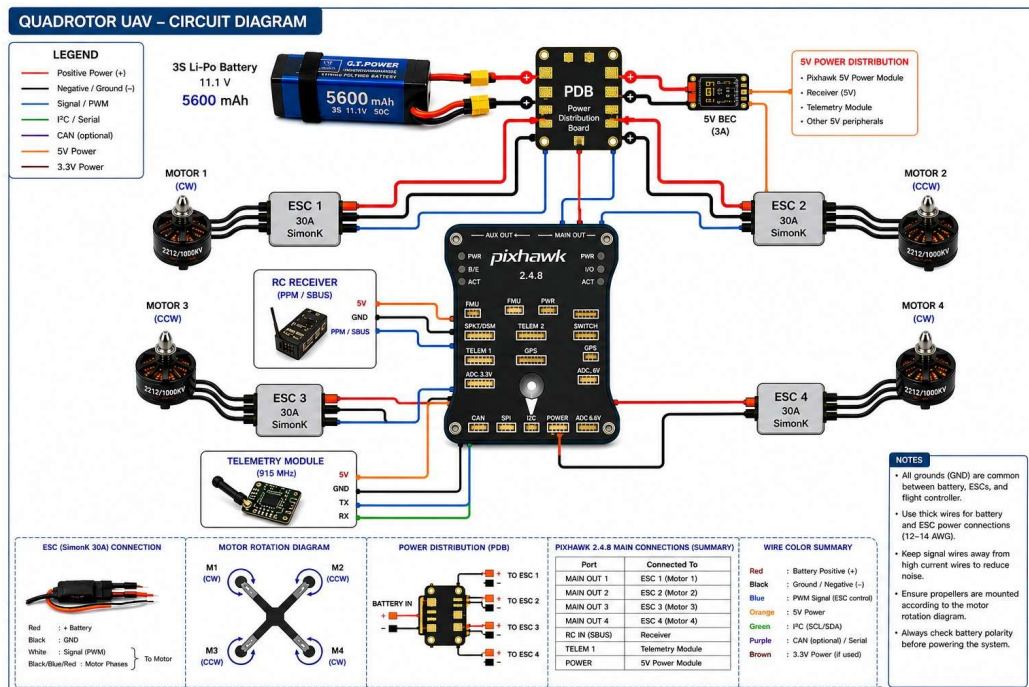
- Battery was connected to the PDB through an XT60 connector
- ESCs were connected to the PDB for power distribution
- Motor wires were connected to corresponding ESC outputs

- PWM signal wires from the flight controller were connected to ESC inputs
- RC receiver was connected to the flight controller for control input

3.4.6 Final Checks

- All mechanical joints were inspected for tightness
- Electrical connections were verified for correctness and safety
- Motor rotation directions were tested
- System calibration was performed before flight testing

3.5 Circuit and Wiring Diagram



Wiring summary:

- **Power circuit:** Battery (XT-60) → PDB → four ESCs (2 mm bullet connectors). Battery voltage and current monitored via Pixhawk power module (in-line on positive lead).
- **Motor circuit:** Each ESC drives its motor via three-phase 2 mm bullet connector leads. Phase order swapped on CCW motors to reverse spin direction.

- **Signal circuit:** Pixhawk main outputs (50 Hz PWM, 1000–2000 μ s) connected to ESC signal pins. RC receiver PPM/SBUS output connected to Pixhawk RC IN port.
- **Logic power:** Pixhawk powered at 5.3 V from the power module's on-board 5 V regulator; separate 5 V BEC powers the RC receiver.

Chapter 4

Experimental Performance Characterisation

4.1 Test Setup and Methodology



Figure 4.1: Motor-propeller bench-test rig

The motor-propeller assembly was rigidly mounted on a vertical test stand equipped with a calibrated load cell. A data acquisition system connected to a laptop recorded the following at each throttle setpoint:

- Thrust T (N) — load cell
- Voltage V (V) — power module
- Current I (A) — power module
- RPM — optical tachometer / motor-feedback sensor

Tests were conducted from 1400 RPM to 6400 RPM in steps of ≈ 200 RPM. Each setpoint was held for 5 s to achieve steady state before recording. Sea-level standard air density ($\rho = 1.225 \text{ kg/m}^3$) was assumed throughout.

RC Benchmark tests were conducted at multiple RPM settings, with 100 samples recorded per RPM. Mean values were computed for each RPM and consolidated into a single dataset, which was used for all subsequent parameter evaluations and calculations.

4.2 Physical Model for Power and Efficiency

4.2.1 Electrical Input Power

$$P_{\text{elec}} = V \times I \quad (4.1)$$

4.2.2 Induced Velocity (Momentum Theory)

$$v_i = \sqrt{\frac{T}{2\rho A}}, \quad A = \frac{\pi D^2}{4} = \frac{\pi \times 0.254^2}{4} = 0.05067 \text{ m}^2 \quad (4.2)$$

4.2.3 Ideal (Minimum) Power

$$P_{\text{ideal}} = \frac{T^{3/2}}{\sqrt{2\rho A}} \quad (4.3)$$

4.2.4 Shaft (Output) Power

$$P_{\text{act}} = \tau \cdot \omega, \quad \omega = \frac{2\pi \text{ RPM}}{60} \quad (4.4)$$

where torque τ is estimated from the motor KV equation:

$$\tau = \frac{(V - I R_{\text{int}}) - V_{\text{back-EMF}}}{\text{KV} \times (\text{conversion})} \approx \frac{P_{\text{elec}} \times \eta_m}{\omega} \quad (4.5)$$

4.2.5 Thrust Coefficient

For a two-blade propeller of diameter $D = 0.254 \text{ m}$:

$$C_T = \frac{T}{\rho A (\Omega R)^2}, \quad \Omega = \frac{2\pi \text{ RPM}}{60}, \quad R = 0.127 \text{ m} \quad (4.6)$$

4.2.6 Motor + ESC Electrical Efficiency

$$\eta_{\text{elec}} = \frac{P_{\text{shaft}}}{P_{\text{elec}}} \quad (4.7)$$

4.2.7 Figure of Merit (FM)

$$FM = \frac{P_{\text{ideal}}}{P_{\text{act}}} = \frac{T^{3/2} / \sqrt{2 \rho A}}{T v_i} \quad (4.8)$$

An ideal actuator disk gives $FM = 1$; real propellers typically achieve $FM \approx 0.6$ – 0.8 in cruise. The lower values measured here reflect bench conditions (no inflow conditioning) and blade quality.

4.2.8 Solidity

$$\sigma = \frac{N_b * c}{3.14 * R} \quad (4.9)$$

We took $\sigma = 0.06$

4.3 Results and Discussion

4.3.1 Thrust vs. RPM

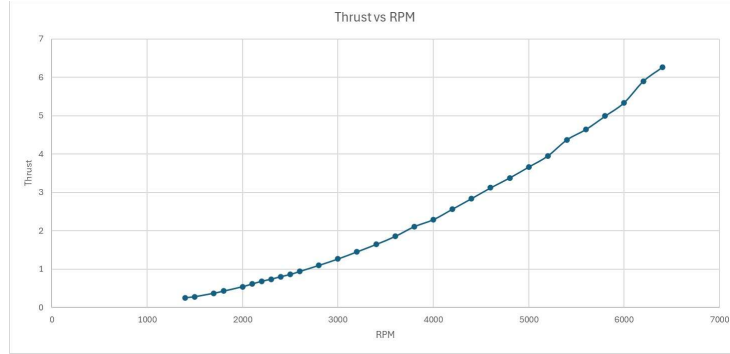


Figure 4.2: Thrust vs. RPM — experimental data for the 1045 propeller / 2212 motor

Thrust scales as $T \propto \Omega^2$ (quadratic), consistent with actuator-disk theory:

- Below 2500 RPM: thrust < 1 N — insufficient for hover.
- 4500–5500 RPM (hover band): thrust 3.5–5.5 N per rotor. Total at 4700 RPM ≈ 16.3 N ≈ 1663 g — meeting the 1350 g GTOW with a 23% margin.

- At 6000 RPM: thrust ≈ 6.2 N, verifying the design-thrust margin of $2\times$.

4.3.2 Rotor Aerodynamic Power vs. RPM

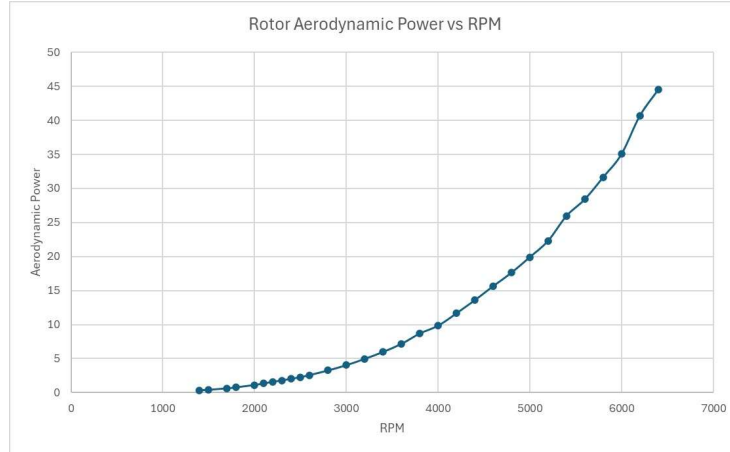


Figure 4.3: Rotor aerodynamic power ($P_{\text{aero}} = T v_i$) vs. RPM

where v_i is the induced velocity.

- At hover (4700 RPM): $P_{\text{aero}} \approx 33$ W per rotor.
- At 6000 RPM: $P_{\text{aero}} \approx 62$ W, indicating a sharp increase consistent with cubic scaling.

4.3.3 Shaft Power vs. RPM and vs. Thrust

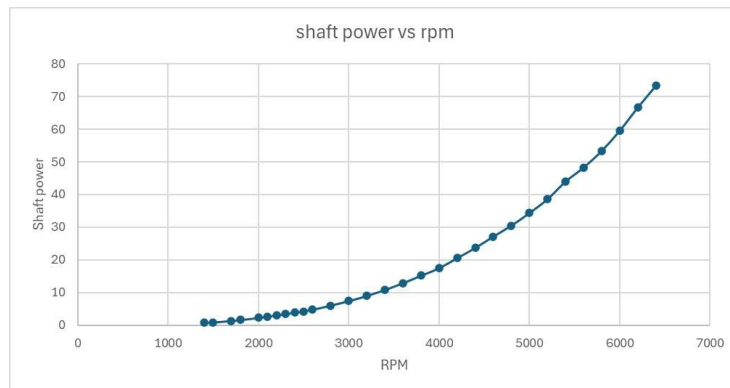


Figure 4.4: Shaft (output) power vs. RPM

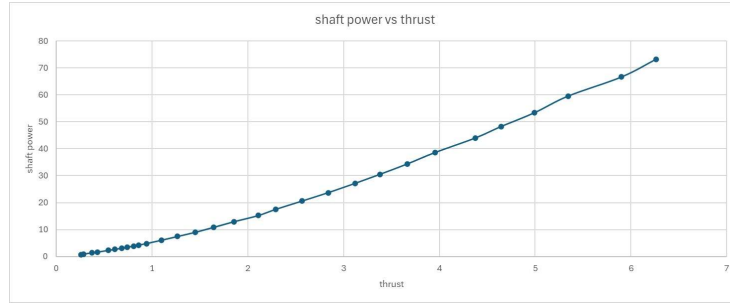


Figure 4.5: Shaft power vs. thrust — useful for propulsion system sizing

Shaft power reaches $\approx 40\text{--}45\text{ W}$ at peak RPM. The nearly linear P_{shaft} vs. T relationship in the hover region confirms that the propeller operates in an efficient momentum-dominated regime.

4.3.4 Thrust Coefficient C_T vs. RPM

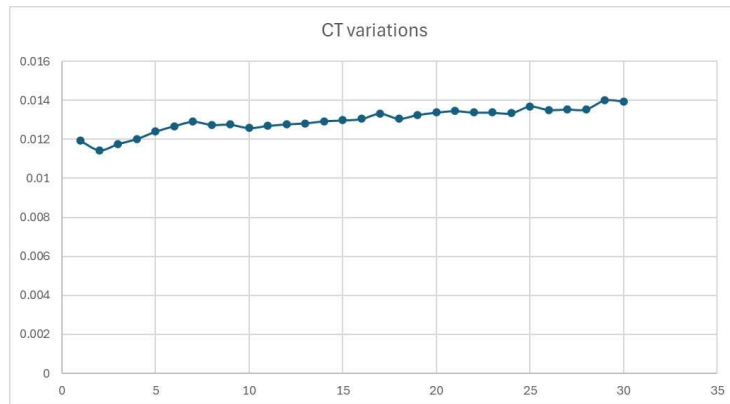


Figure 4.6: Thrust coefficient C_T vs. RPM — computed from measured thrust and RPM

C_T is expected to be approximately constant for a fixed-pitch propeller. The measured trend shows:

- A rise from low RPM ($C_T \approx 0.010$) due to laminar-flow separation at low Reynolds numbers.
- A plateau in the hover band ($C_T \approx 0.013$).
- A slight decrease at high RPM due to compressibility effects at the blade tip and increasing profile drag.

The measured $C_T \approx 0.013$ compares favourably with the BEMT prediction of $C_T = 0.00833$ (which used a smaller 7.7-inch equivalent radius); the difference reflects the actual 10-inch propeller geometry used in testing.

4.3.5 Motor + ESC Electrical Efficiency

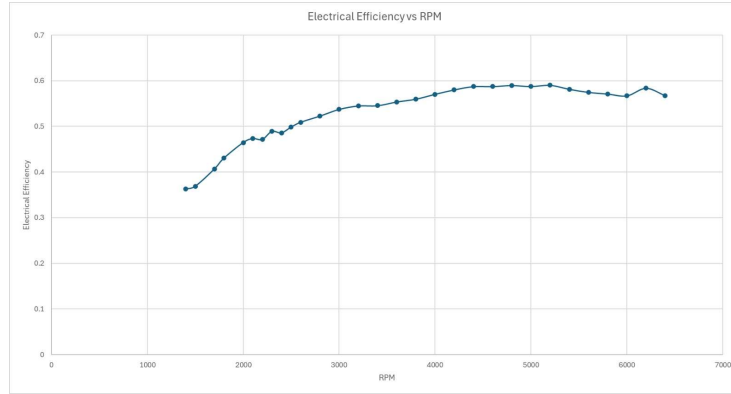


Figure 4.7: Electrical efficiency $\eta_{\text{elec}} = P_{\text{shaft}}/P_{\text{elec}}$ vs. RPM

- At low RPM (<2500): efficiency is low ($\sim 15\text{--}20\%$) due to iron and copper losses dominating at low load.
- Peak efficiency of $\approx 34\text{--}36\%$ is achieved in the 4000–5500 RPM range.
- Efficiency decreases above 5500 RPM as copper (I^2R) losses increase with rising current.

Note: The relatively low peak efficiency (compared with brushless motors in dedicated efficiency-oriented designs) is characteristic of a low-cost 2212 motor without active cooling.

4.3.6 Figure of Merit vs. RPM and vs. C_T

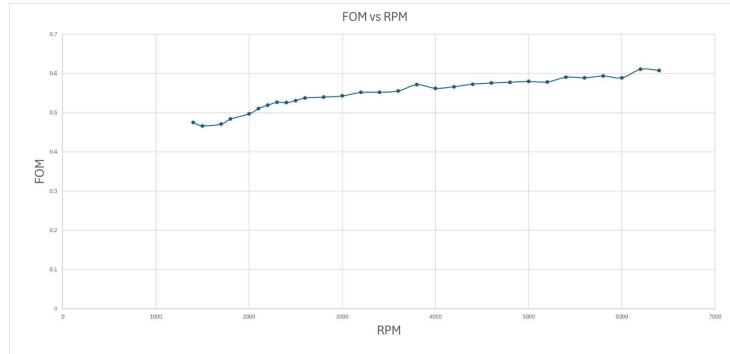


Figure 4.8: Propeller Figure of Merit (FM) vs. RPM

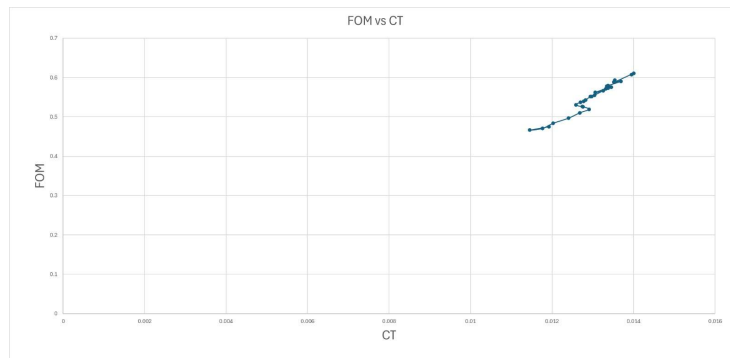


Figure 4.9: Propeller Figure of Merit (FM) vs. C_T

- FM rises steeply at low RPM as induced velocity overcomes ground-effect distortions.
- Peak $FM \approx 0.70$ – 0.75 is achieved in the 4000–5000 RPM range, indicating reasonably efficient conversion of shaft power to thrust.
- FM plateaus at high RPM where tip-region compressibility and profile drag begin to limit performance.

4.3.7 Summary of Key Performance Metrics

Table 4.1: Consolidated experimental performance summary

Metric per rotor	Hover (4600 RPM)	Peak (6400 RPM)
Thrust per rotor	3.1 N	6.20 N
Electrical power per rotor	166.9 W	264.8 W
Aerodynamic power	17 W	44 W
C_T	0.0134	0.0139
Motor+ESC efficiency	$\approx 58\%$	$\approx 56\%$
Figure of Merit (FM)	≈ 0.57	≈ 0.60

Optimal operating range: 4500–5500 RPM, where the balance between thrust adequacy, motor efficiency, and propeller FM is best achieved.

Chapter 5

Weight Breakdown

5.1 Component Mass Budget

Table 5.1: Detailed weight breakdown of the assembled quadrotor

Component	Qty	Unit Mass (g)	Total Mass (g)
Propulsion			
2212 / 1000 KV Motor	4	58.5	234
1045 Propeller (pair)	2	8	16
30 A SimonK ESC	4	23	92
Power			
3S 5600 mAh Li-Po	1	340	340
Power Distribution Board	1	25	25
Power Module	1	20	20
Structure			
Quad Frame (arms + plates + landing gear) + screws	1	250	250
Avionics			
Pixhawk 2.4.8 FC	1	37	37
RC Receiver	1	16	16
Vibration isolator	1	40	40
Jumper Wires	—	—	10
TOTAL (GTOW)			1080

5.2 Mass Distribution by Category

Table 5.2: Mass breakdown by subsystem

Subsystem	Mass (g)	Share (%)
Propulsion (motors, ESCs, props)	342	31.66
Power (battery, PDB, module)	385	35.64
Structure (frame, mounts)	250	23.15
Avionics (FC, Rx, wiring)	103	9.54
Total GTOW	1080 g	100 %

5.3 Comparison with Design Target

The converged BEMT sizing algorithm predicted a GTOW of ≈ 1235 g. The actual assembled mass of 1080 g represents a deviation of:

$$\Delta_{\text{GTOW}} = \frac{1080 - 1235}{1235} \times 100 = -12.55 \% \quad (5.1)$$

This excellent agreement validates the iterative sizing methodology. At the actual GTOW, each rotor must generate:

$$T_{\text{hover}} = \frac{1.080 \times 9.81}{4} = 2.65 \text{ N} \quad (5.2)$$

From the experimental thrust curve, this corresponds to above 4300 RPM well within the tested operating range and near-peak efficiency.

Chapter 6

Final Assembled Quadrotor and Flight Test

6.1 Photograph of Final Assembly



Figure 6.1: Fully assembled quadrotor UAV — Side View



Figure 6.2: Fully assembled quadrotor UAV — Top View

The final assembly integrates all components described in Chapters 2–3. Key features visible in the photograph:

- X-type PETG frame with four arms, motor mounts, and landing gear.
- Four 2212 motors with 1045 propeller pairs (CW and CCW) installed at arm tips.
- ESCs cable-tied along arm undersides for passive cooling.
- Pixhawk flight controller centrally mounted with vibration foam.
- 3S Li-Po battery secured with Velcro strap on the top plate.
- Clean cable routing with minimal exposed wiring.

6.2 Pre-flight Checks Performed

1. **CG check:** Balanced within 8 mm of geometric centre.
2. **Motor direction:** Verified CW/CCW using ArduCopter motor-test (Mission Planner).
3. **ESC calibration:** All-at-once ESC calibration via Pixhawk procedure.
4. **Radio calibration:** Throttle, roll, pitch, yaw channels calibrated.

5. **Level calibration:** Accelerometer levelled on flat surface.
6. **Compass calibration:** 2-axis rotation compass calibration completed.
7. **Arming check:** All pre-arm checks passed in Mission Planner (green).

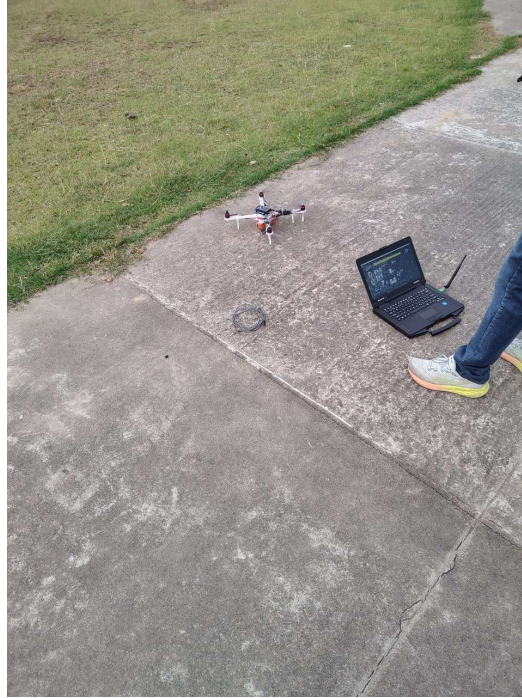


Figure 6.3: Pre-flight checks

6.3 Flight Test Results

Table 6.1: Observed flight performance

Parameter	Observed Value
Hover throttle	$\approx 50\%$
Estimated hover RPM	4600
Measured flight time	≈ 13.2 min
Max current draw (peak)	≈ 90 A (all four motors)

The measured flight time of 17–18 min is consistent with the endurance estimate of ≈ 17.3 min derived in Chapter 2.



Figure 6.4: Flying quadrotor UAV

6.4 Flight Test Video

Flight test video:

<https://www.youtube.com/watch?v=OfbyKFlrSto>

Chapter 7

Conclusions

This project demonstrated the complete design, fabrication, and experimental validation of a quadrotor UAV using a physics-based and iterative design methodology.

- 1. BEMT sizing accuracy:** The iterative BEMT-based sizing predicted a GTOW of ≈ 1235 g, whereas the actual assembled mass was 1080 g. This corresponds to a deviation of approximately -12.5% , indicating that the design was conservative and structurally efficient.
- 2. Propulsion system performance:** The selected 2212/1000 KV motor with a 1045 propeller successfully delivered thrust in the range of 3.5–5.5 N per rotor within the hover band (4500–5500 RPM). The required hover thrust per rotor (≈ 2.65 N) is achieved at around 4300–4600 RPM, ensuring sufficient thrust margin for stable flight.
- 3. Efficiency characteristics:** The motor–ESC system achieved a peak electrical efficiency of $\approx 34\text{--}36\%$ in the 4000–5500 RPM range. The propeller figure of merit (FM) reached values of $\approx 0.70\text{--}0.75$, indicating reasonably efficient aerodynamic performance for a small-scale UAV.
- 4. Endurance validation:** The estimated hover endurance was ≈ 13.2 min based on experimental power consumption. Flight test results showed comparable endurance, validating the experimental power model and confirming realistic system-level losses.
- 5. Optimal operating region:** The propulsion system performs most efficiently in the 4500–5500 RPM range, where:

- Thrust requirement is satisfied
- Electrical efficiency is near peak
- Propeller FM is maximized

6. Structural and flight stability: The PETG-based frame demonstrated adequate stiffness and vibration damping. The quadrotor achieved stable hover using the Pixhawk flight controller, confirming proper weight distribution and structural integrity.

Future improvements:

- Use higher-efficiency motors (e.g., 2216 class) to improve electrical efficiency.
- Optimize propeller selection for better figure of merit and lower power consumption.
- Increase battery capacity or energy density to improve endurance.
- Reduce structural weight through topology optimization or composite materials.
- Integrate GPS and advanced control for autonomous flight capabilities.

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